

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

STG310-R1

Building a globally replicated and resilient Amazon S3 architecture

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Agenda

Multi-Region data storage

Amazon S3 Multi-Region Access Points

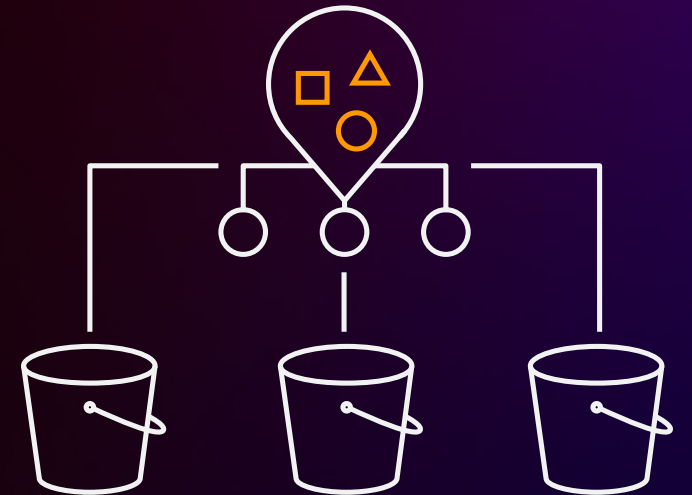
Failover controls

NEW

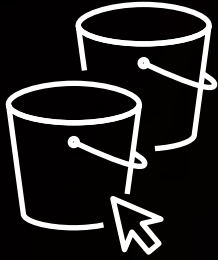
Demo

Cost optimization

Best practices



Why store your data in multiple AWS Regions?



Resilience

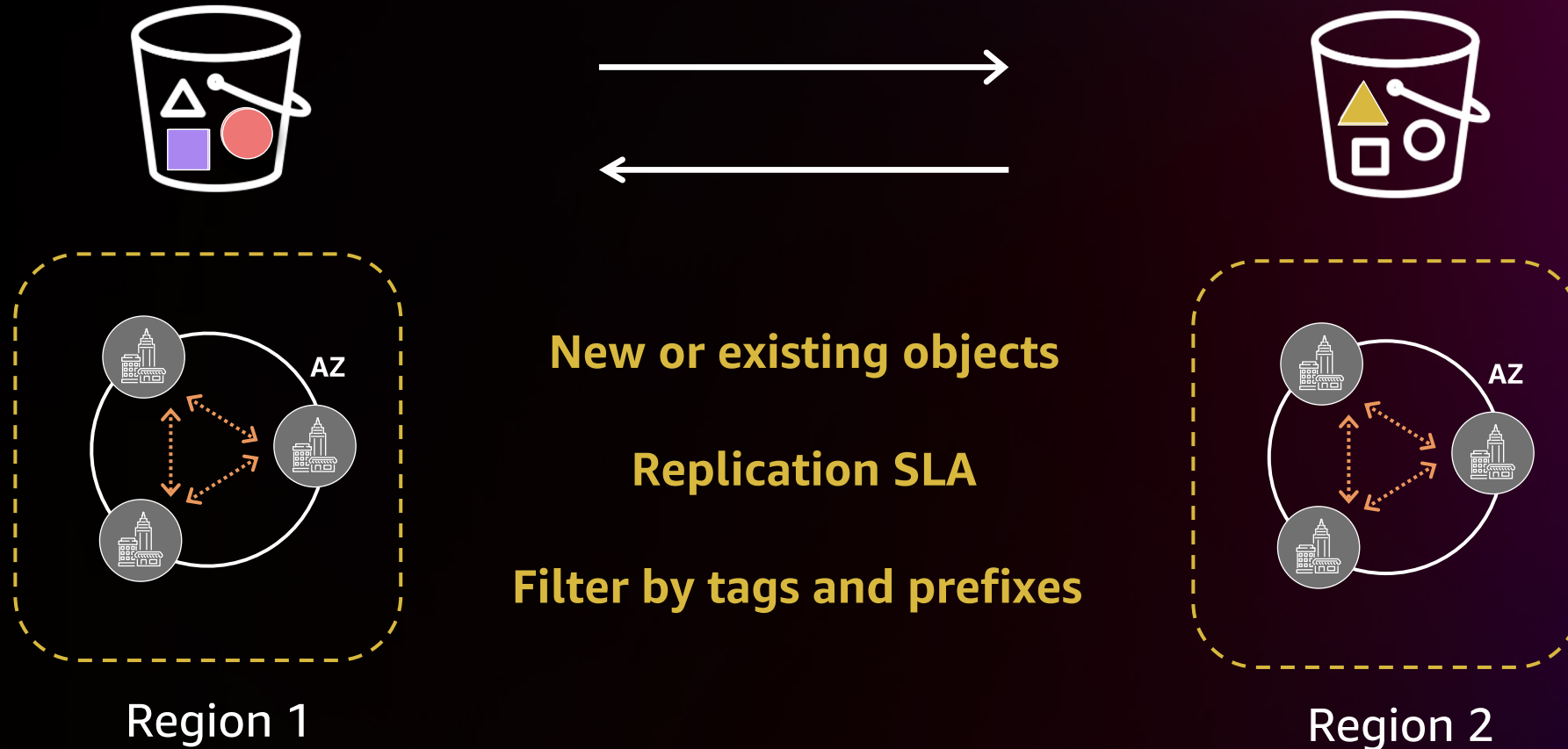


Compliance

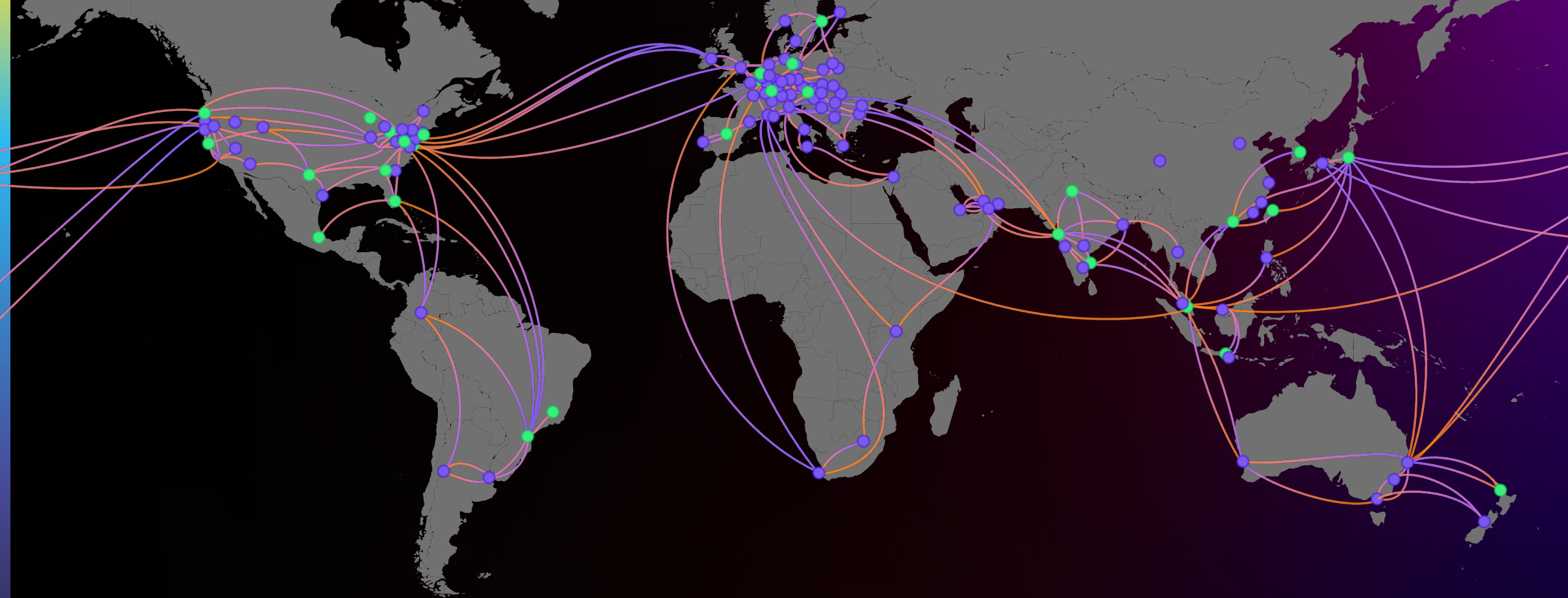


Performance

Amazon S3 Cross-Region Replication



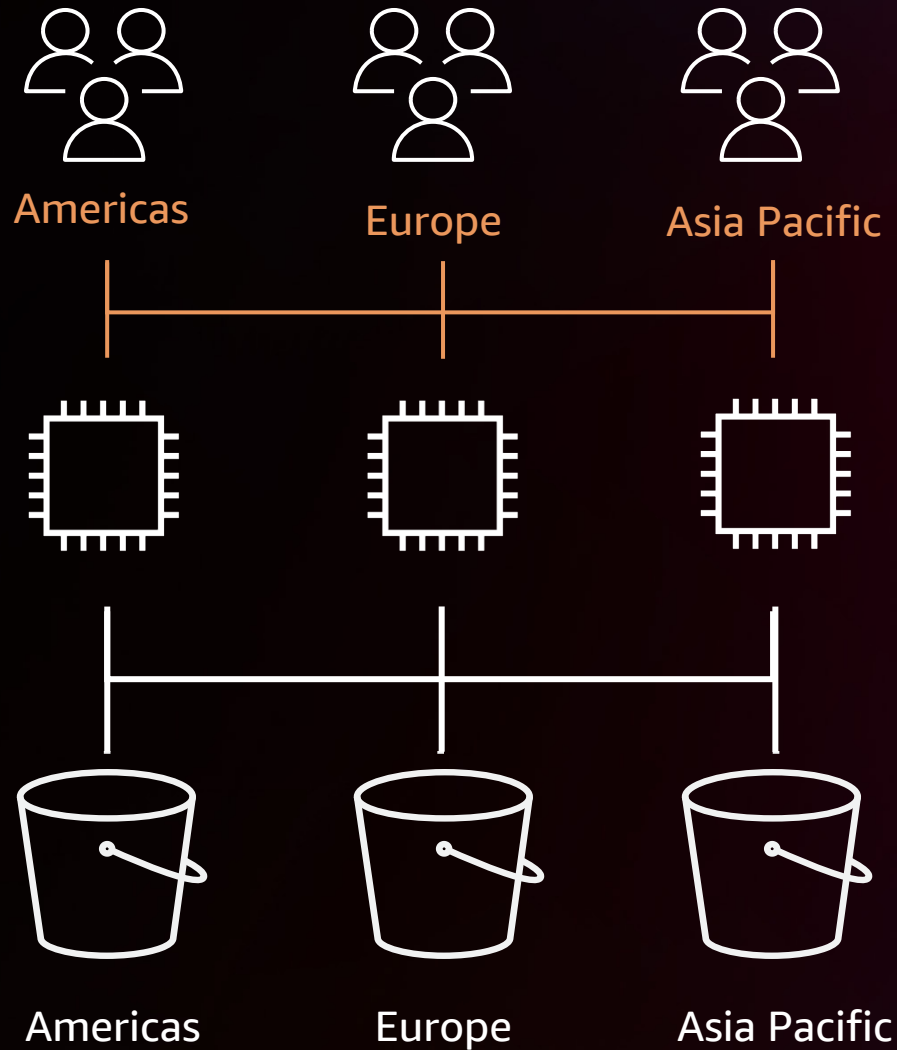
Global communications



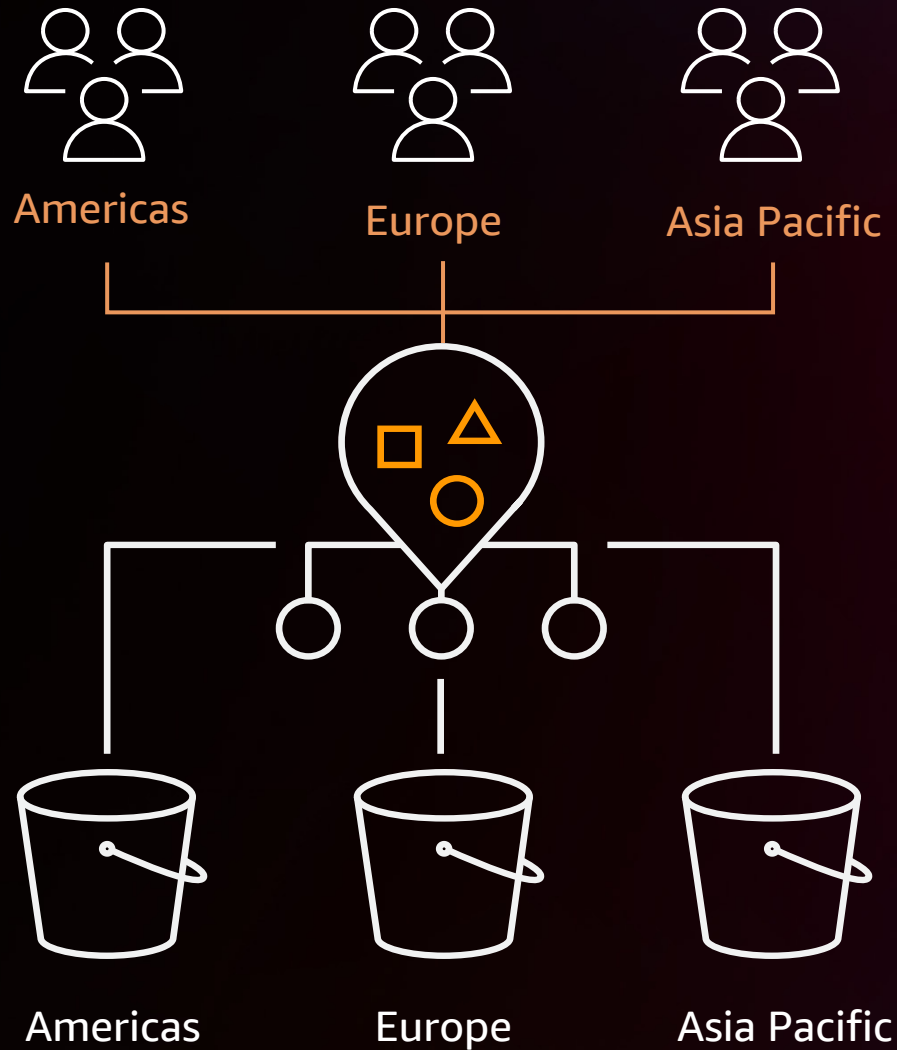
Communications challenges



S3 Multi-Region storage



S3 Multi-Region Access Points



Connecting to your global endpoint

Cross-Region VPC
via AWS PrivateLink
(no VPC peering)



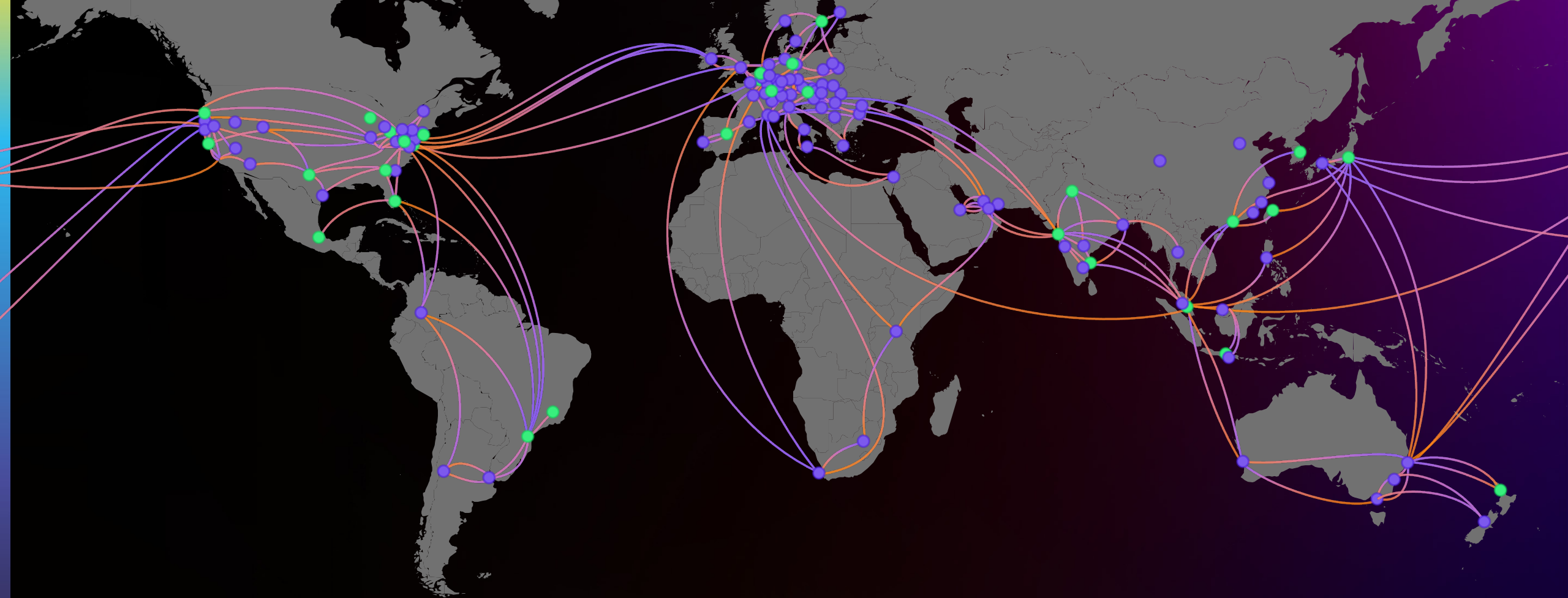
On premises
via AWS PrivateLink



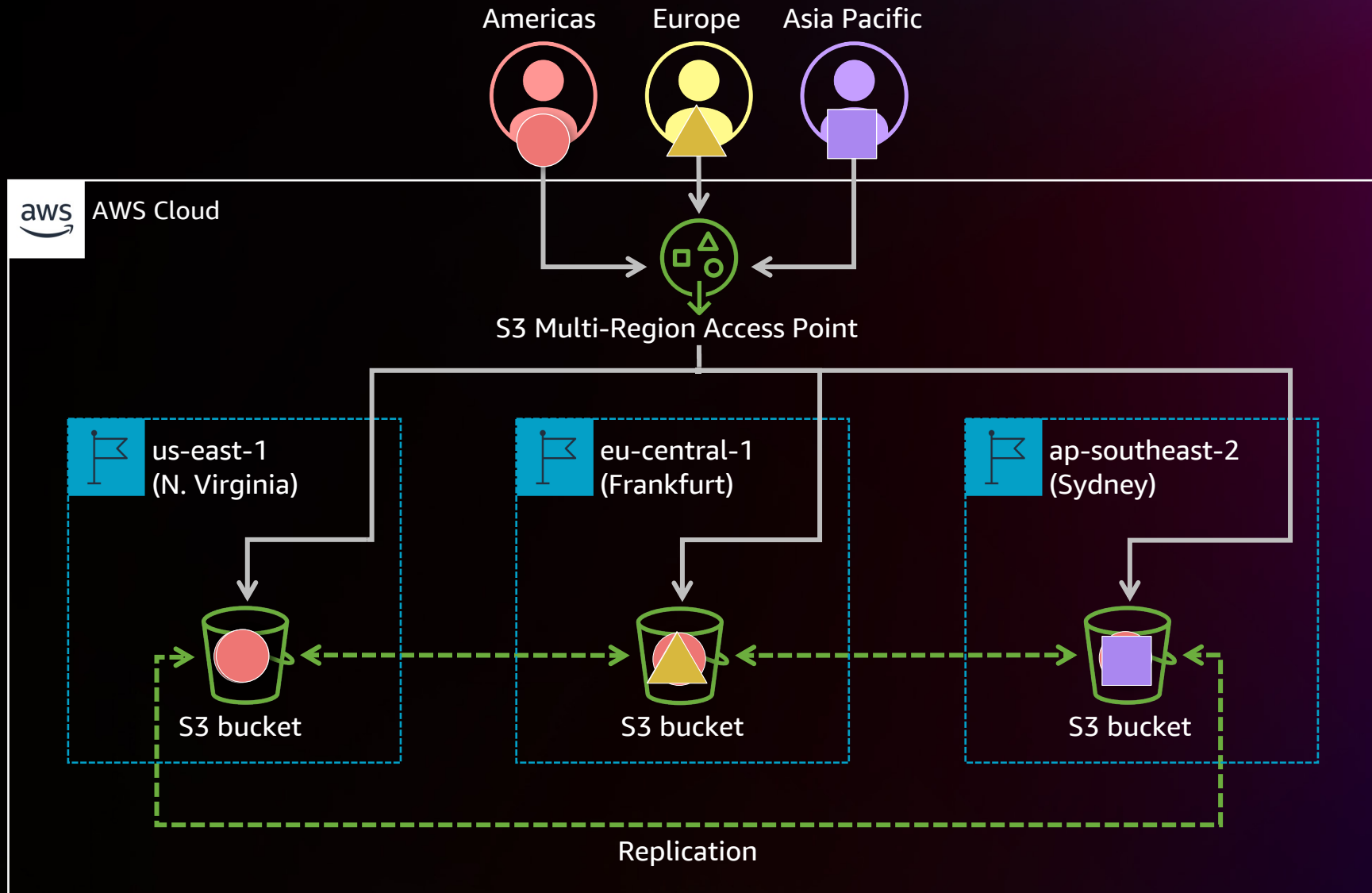
Internet acceleration



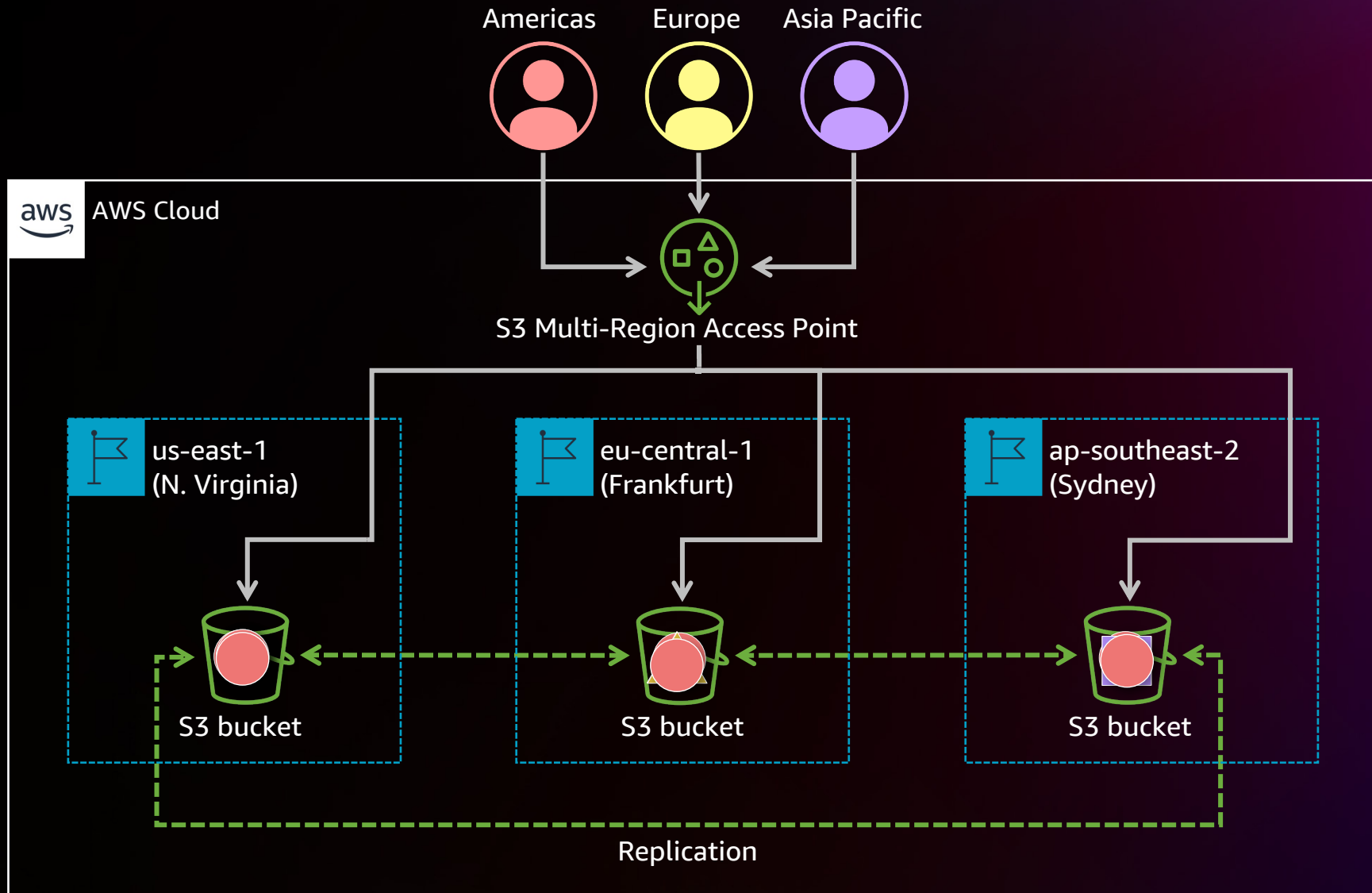
Global communications



Active-Active global storage

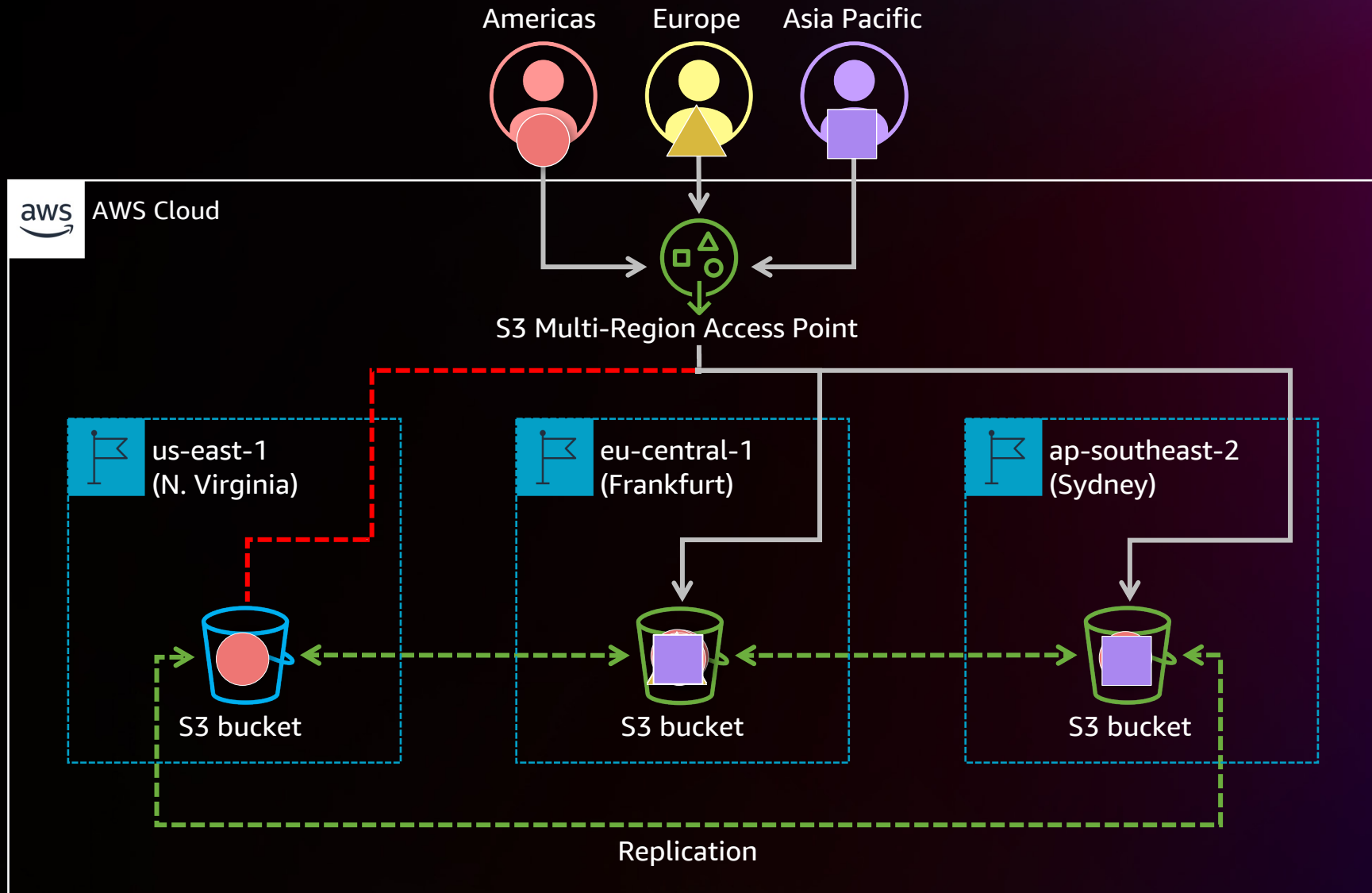


Active-Active global storage



S3 Multi-Region Access Points failover controls

NEW



Whiteboard



Multi-Region failover controls CLI + APIs

CLI

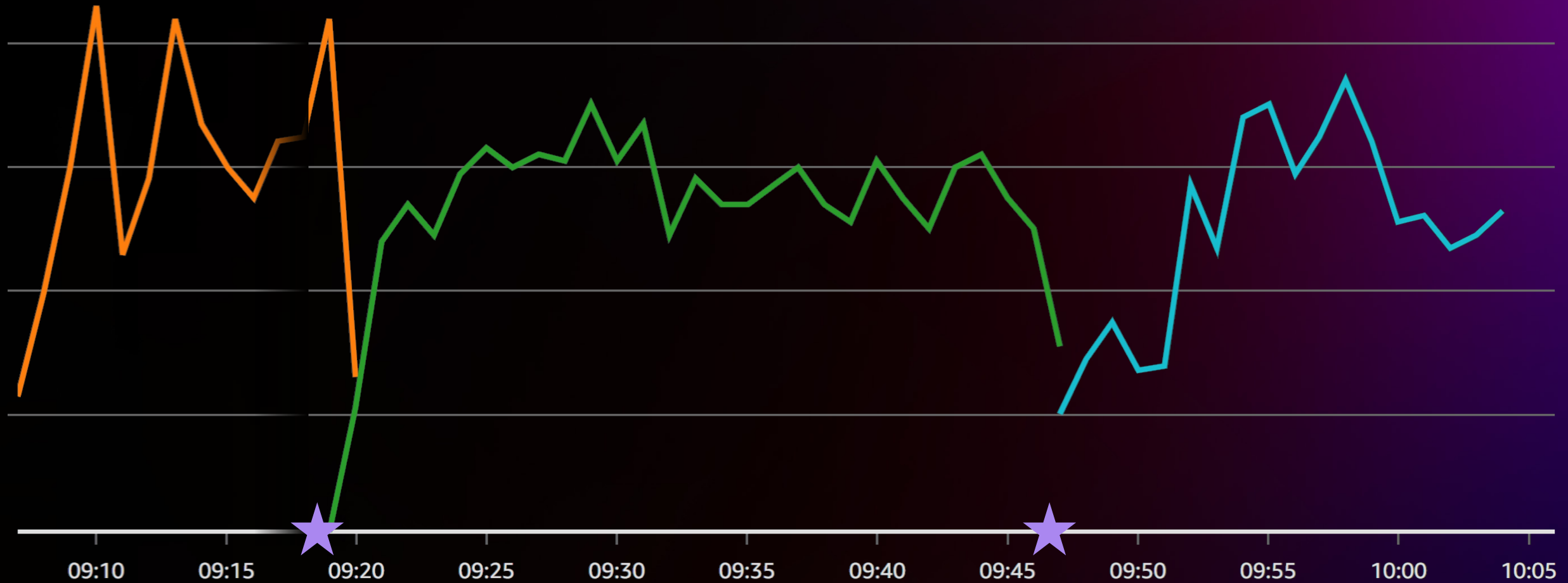
```
aws s3control submit-multi-region-access-point-routes \
--region ap-southeast-2 \
--account-id AccountId \
--mrap mrap_ARN \
--route-updates Bucket=Bucket_1,TrafficDialPercentage=100 \
                 Region=Region_2,TrafficDialPercentage=0
```

SDK

```
PATCH /v20180820/mrap/instances/mrap+/routes HTTP/1.1
Host: s3-control.amazonaws.com
x-amz-account-id: AccountId
<?xml version="1.0" encoding="UTF-8"?>
<SubmitMultiRegionAccessPointRoutesRequest
xmlns="http://awss3control.amazonaws.com/doc/2018-08-20/">
  <RouteUpdates>
    <Route>
      <Bucket>string</Bucket>
      <Region>string</Region>
      <TrafficDialPercentage>integer</TrafficDialPercentage>
    </Route>
  </RouteUpdates>
</SubmitMultiRegionAccessPointRoutesRequest>
```

Monitoring failover

AMAZON CLOUDWATCH GRAPH OF GET REQUESTS TO EACH S3 BUCKET



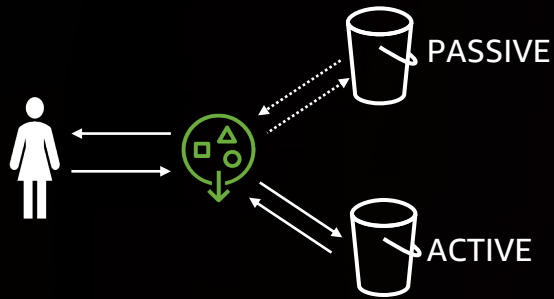
S3 buckets in `ap-southeast-2`, `us-east-1`, and `us-west-2` during routing updates

Whiteboard



Combining Amazon S3 features

S3 Multi-Region Access Points



Automatic or manual failover

S3 Intelligent-Tiering



Automatic cost optimization

No **retrieval** fees

S3 Replication Time Control

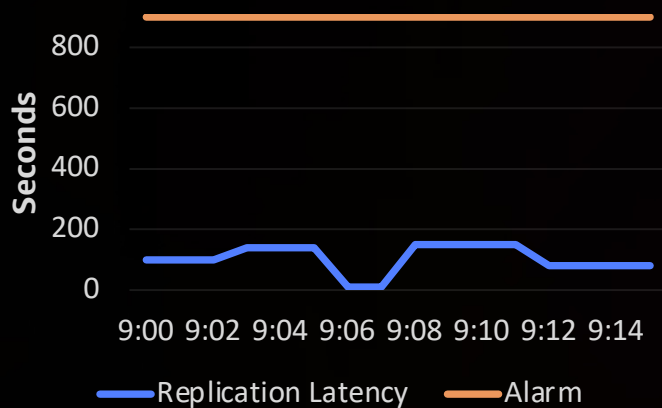


Replication **SLA**

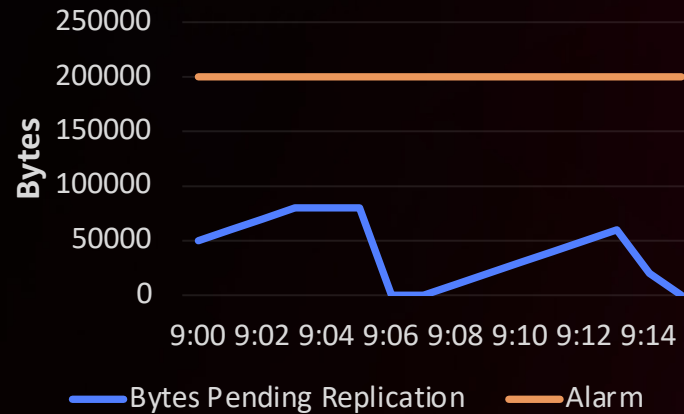
Amazon S3 Replication Time Control

DESIGNED TO REPLICATE 99.99% OF OBJECTS WITHIN 15 MINUTES

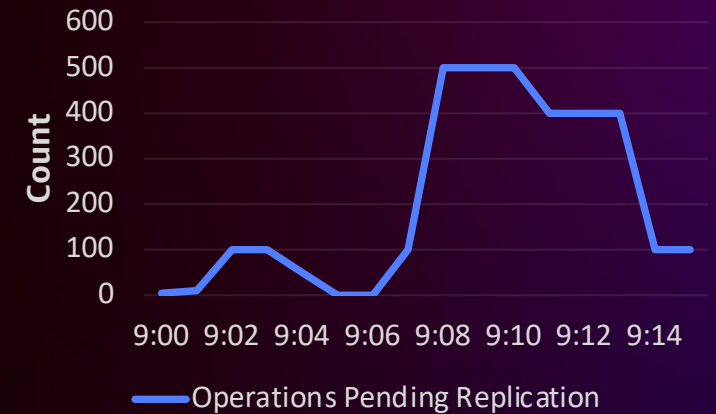
Monitor your replication with **CloudWatch metrics**



Replication latency



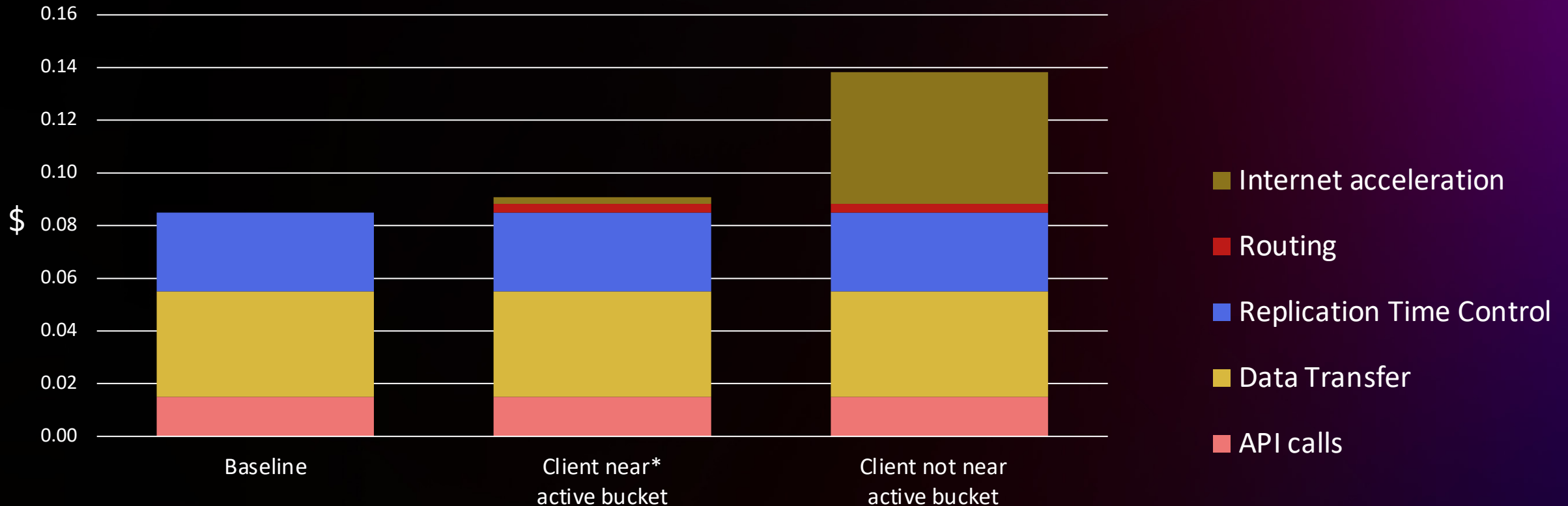
Bytes pending replication



Operations pending replication

Incremental cost of Multi-Region Access Points

Cost to **write 1 GB** of 1 MB objects, including replication to 2 other buckets

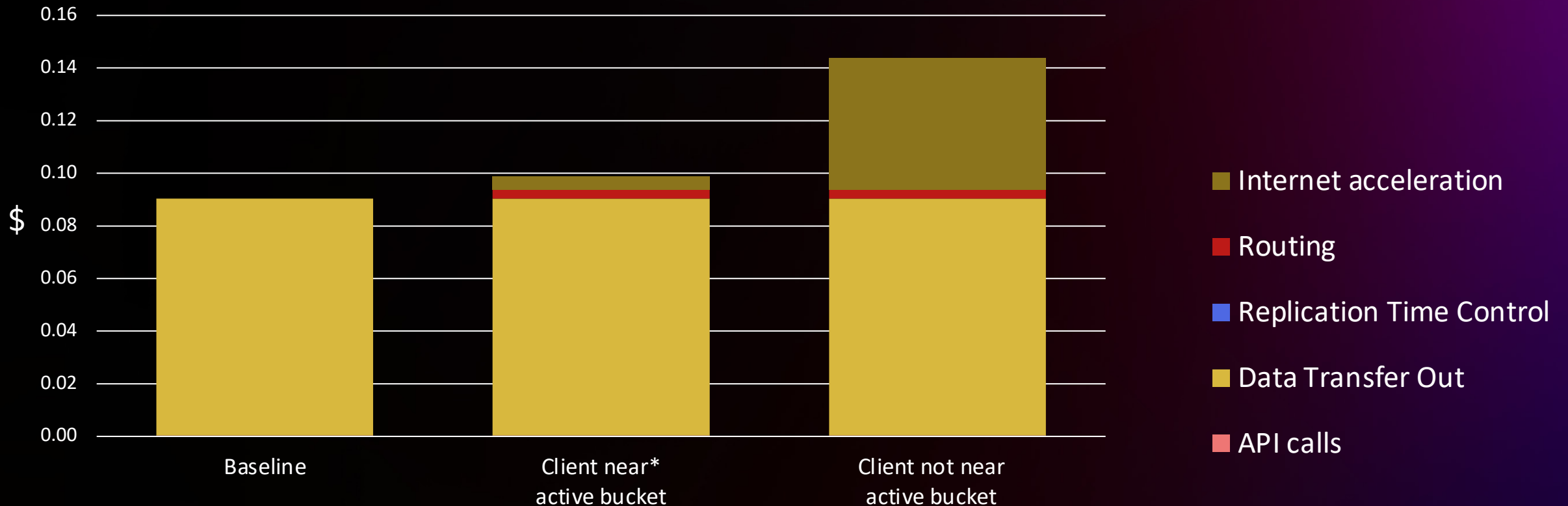


API calls based on \$0.05 per 1,000 PUTs. Acceleration pricing for sources or destinations within North America or Europe. Data transfer based on \$0.02/GB.

*i.e., client and bucket are both in North America, South America, Europe, or Asia Pacific.

Incremental cost of Multi-Region Access Points

Cost to **read 1 GB** of 1 MB objects using an internet-based client



API calls based on \$0.004 per 1000 GETs. Acceleration valid for sources or destinations within North America or Europe. Data transfer out pricing in <10 TB/mth band.

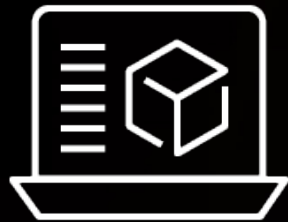
*i.e., client and bucket are both in North America, South America, Europe, or Asia Pacific.

Getting started with Multi-Region Access Points

1. Update SDKs to the latest version
2. Existing or new buckets in the same account, in up to 17 Regions
3. A bucket can be used behind more than one Multi-Region Access Point
4. Use the **.s3-global.accesspoint** VPC interface endpoint
5. Understand what data access operations are supported
6. Leverage Amazon CloudWatch to monitor shifting S3 traffic after a failover

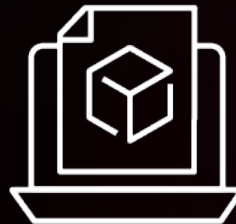
Continue your AWS storage learning

**Build a
learning plan**



Set your AWS storage
learning plans via
AWS Skill Builder

**Increase your
knowledge**



Use our **ramp-up guides**
to build your storage
knowledge

**Earn AWS
Storage badges**



Demonstrate your
knowledge by achieving
digital badges

aws.training/storage



Thank you!

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